

## A.2 Influence on Local Neighbors in GloVe Space

| Word         | Neighbors before                     | Neighbors after                           |
|--------------|--------------------------------------|-------------------------------------------|
| order        | orders, ordering, purchase           | orders, ordering, ordered                 |
| crack        | keygen, cracks, torrent              | keygen, cracks, warez                     |
| craigslist   | ebay, craigslist, ads                | ebay, craigslist, freecycle               |
| populations  | population, species, communities     | population, species, habitats             |
| epub         | ebook, mobi, pdf                     | mobi, ebook, kindle                       |
| finals       | semifinals, playoffs, championship   | semifinals, semifinal, quarterfinals      |
| installed    | install, installing, installation    | install, installing, installs             |
| identifiable | disclose, identify, identifying      | disclose, pii, distinguishable            |
| photographs  | photograph, photos, images           | photograph, images, photos                |
| ta           | si, tu, ti                           | que, bien, ele                            |
| couch        | sofa, sitting, bed                   | sofa, couches, loveseat                   |
| cooler       | coolers, cooling, warmer             | coolers, cooling, warmer                  |
| becky        | debbie, kathy, julie                 | debbie, steph, jen                        |
| appreciated  | appreciate, greatly, thanks          | appreciate, muchly, thanks                |
| negotiation  | negotiating, negotiations, mediation | negotiating, negotiations, mediation      |
| initial      | subsequent, prior, following         | intial, inital, subsequent                |
| chloe        | chanel, emma, lauren                 | chlo  , chanel, handbags                  |
| filipino     | pinoy, filipinos, philippine         | filipinos, pinoy, tagalog                 |
| relying      | rely, relied, relies                 | rely, relied, relies                      |
| perpetual    | eternal, continual, irrevocable      | irrevocable, datejust, perpetuity         |
| himself      | him, herself, his                    | herself, oneself, he                      |
| seaside      | beach, beachside, picturesque        | beachside, idyllic, seafront              |
| measure      | measures, measuring, measured        | measures, measuring, measured             |
| yorkshire    | staffordshire, leeds, lancashire     | staffordshire, dales, lancashire          |
| merchandise  | goods, items, apparel                | goods, items, merchandize                 |
| sub          | subs, k, def                         | subs, subbed, svs                         |
| tones        | tone, hues, muted                    | tone, polyphonic, muted                   |
| therapist    | therapists, psychologist, therapy    | therapists, physiotherapist, psychologist |
| leaned       | sighed, smiled, glanced              | leant, leaning, sighed                    |
| tho          | nnd, cuz, tlie                       | nnd, tlio, tlie                           |
| lawyers      | attorneys, lawyer, attorney          | attorneys, lawyer, attorney               |
| compile      | compiling, compiler, compiles        | compiling, compiler, compiles             |
| chord        | chords, progressions, guitar         | chords, progressions, voicings            |
| aims         | aim, aimed, aiming                   | aim, aimed, aiming                        |
| ensure       | ensuring, assure, ensures            | ensuring, ensures, assure                 |
| aerospace    | aviation, engineering, automotive    | aeronautics, aviation, aeronautical       |
| clubhouse    | pool, playground, amenities          | clubhouses, pool, playground              |
| locking      | lock, locks, latch                   | lock, locks, latch                        |
| reign        | reigns, emperor, throne              | reigns, reigned, emperor                  |
| vulnerable   | susceptible, fragile, affected       | susceptible, vunerable, fragile           |

Table 1: 3-nearest words before and after the INLP projection

Table 1 above presents the results of word-embeddings similarity test mentioned in 6.1. This table lists the top 3-nearest neighbors of sampled words from GloVe, before and after the INLP process. It is evident that INLP does not alter the neighbors of the random sample in a detrimental way.

### A.3 Quantitative Influence of Gender Debiasing on Glove Embeddings

In Appendix A.2 we provide a sample of words to qualitatively evaluate the influence of INLP on semantic similarity in Glove word embeddings (Section 6.1). We observe minimal change to the nearest neighbors. To complement this measure, we use a quantitative measure: measuring performance on established word-similarity tests, for the original Glove embeddings, and for the debiased ones. Those tests measure correlation between cosine similarity in embedding space and human judgements of similarity. Concretely, we test the embeddings similarities using three dataset, which contain four similarity tests that measure similarity or relatedness between words. We use the following datasets: SimLex999 (Hill et al., 2015), WordSim353 (Agirre et al., 2009) which contain two evaluations, on words similarity and relatedness and finally on Mturk-771 (Halawi et al., 2012).

The test sets are composed of word pairs, where each pair was annotated by humans to give a similarity or relatedness score. To evaluate a model against such data, each pair is given a score (in the case of word embedding, cosine similarity) and then we calculate Spearman correlation between all the score pairs. The results on the regular Glove embeddings before and after the gender debiasing are presented in Table 3. We observe a major improvements across all evaluation sets after the projection: between 0.044 to 0.116 points.

This major difference in performance is rather surprising. It is not clear how to interpret the positive influence on correlation with human judgements. This puzzle is further compounded by the fact the projection reduces the rank of the embedding spaces, and by definition induces loss of information. We hypothesize that many of the words in the embedding space contain a significant gender component, which is not correlated with humans judgements of similarity. While intriguing, testing this hypothesis is beyond the scope of this work, and we leave the more rigorous answer to a future

work.

### A.4 Influence on Local Neighbors of Surnames Representations in Glove Space

| Word      | Neighbors before          | Neighbors after                        |
|-----------|---------------------------|----------------------------------------|
| ruth      | helen, esther, margaret   | eting, esther, gehrig                  |
| charlotte | raleigh, nc, atlanta      | raleigh, greensboro, nc                |
| abigail   | hannah, lydia, eliza      | hannah, phebe, josiah                  |
| sophie    | julia, marie, lucy        | moone, bextor, marceau                 |
| nichole   | nicole, kimberly, kayla   | nicole, mya, heiress                   |
| emma      | emily, lucy, sarah        | grint, frain, watson                   |
| olivia    | emma, rachel, kate        | mun, thirby, wilde                     |
| ava       | devine, zoe, isabella     | viticulatural, devine, appellation     |
| isabella  | sophia, josephine, isabel | rossellini, beeton, ferdinand          |
| sophia    | anna, lydia, julia        | hagia, antipolis, topkapi              |
| mia       | bella, mamma, mama        | bangg, mamma, culpa                    |
| amelia    | earhart, lousia, caroline | earhart, fernandina, bedelia           |
| james     | john, william, thomas     | jassie, nightfire, perse               |
| john      | james, william, paul      | deere, scatman, betjeman               |
| robert    | richard, william, james   | pattinson, mccammon, blacksportsonline |
| michael   | david, mike, brian        | micheal, franti, moorcock              |
| william   | henry, edward, james      | edward, henry, sir                     |
| david     | stephen, richard, michael | bisbal, magen, sylvian                 |
| richard   | robert, william, david    | clayderman, brautigan, rorty           |
| joseph    | francis, charles, thomas  | joesph, dreamcoat, abboud              |
| thomas    | james, william, john      | szasz, deshaun, tomy                   |
| ariel     | sharon, alexis, hanna     | peterpan, mermaid, cinderella          |
| mike      | brian, chris, dave        | mignola, birbiglia, dave               |

Table 2: 3-nearest words before and after the INLP projection, for surnames

The results in Table 1 suggest that, as expected, the projection has little influence on the lexical semantics of unbiased words, as measured by their closest neighbors in embedding space. But how does the projection influence inherently gendered words? Table 2 contains the closest-neighbors to the Glove representations of gendered surnames, before and after the projection. We observe an interesting tendency to move from neighbors which are other gendered surnames, towards family names, which are by definition gender-neutral (for instance, the closest neighbor of “Robert” changes from “Richard” to “Pattinson”). Another interesting tendency is to move towards place names bearing a connection to that surnames (For instance, the closest neighbor of “Sophia” changes to “Hagia”). At the same time, some gendered surnames remain close neighbors even after the projection.

### A.5 Performance and “Fair Classification” as a Function of INLP Iterations

In Section 6.2 where we compare the accuracy and TPR-Gap before and after using INLP for a certain amount of iterations. The number of iterations chosen is somehow arbitrary, but we emphasize that this can be controlled for as the number of iterations used with INLP. By sacrificing the main task performance, one can improve the TPR-Gap

| Eval             | Before | After        |
|------------------|--------|--------------|
| SimLex999        | 0.373  | <b>0.489</b> |
| WordSim353 - Sim | 0.695  | <b>0.799</b> |
| WordSim353 - Rel | 0.599  | <b>0.698</b> |
| Mturk-771        | 0.684  | <b>0.728</b> |

Table 3: Word similarity scores on Glove embeddings, before and after INLP. The scores are the Spearman correlation coefficient between the similarity scores.

of their model. In Figure 1 we detail these trade-offs for the 0.8 ratio, where the original TPR-Gap originally is the highest.

We note that the performance is minimally damaged for the first 180 iterations, while the TPR-Gap improves greatly, after-which, both metric account for larger drops. Using this trade-off, one can decide how much performance they are willing to sacrifice in order to get a less biased model.

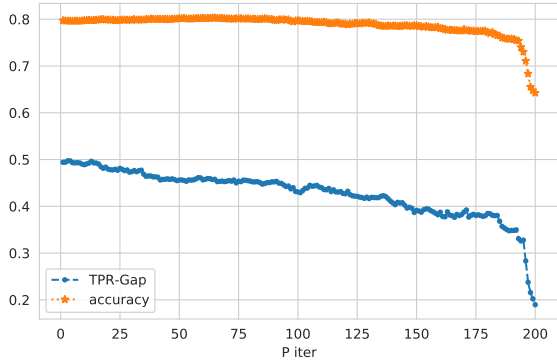


Figure 1

## A.6 Biographies dataset: Words Most-Associated with Gender

### A.6.1 Bag-of-Words Model

In this section, we present the raw results of the experiment aimed to assess the influence of INLP on specific words, under the bag-of-words model, for the biographies experiments (Section 6.3.1).

Table 4 lists the words most influenced by INLP projection (on average over all professions) after the debiasing procedure explained in Section 6.3.

Figure 2 presents the relative change of biased word for each profession, compared to a random sample.

| Most Changed Words                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ms., mr., his, her, he, she, mrs., specializes, english, practices, ', him, spanish, speaks, with, affiliated, and, medicine, ms, state, #, the, medical, michael, in, residency, at, of, psychology, dr., 's, law, research, practice, about, where, business, education, 5, -, is, first, women, america, insurance, more, john, university, location, ph.d., surgery, (, mental, ), that, engineering, graduated, language, bs, litigation, collection, united, 1, graduate, humana, cpas, cancer, npi, completed, 10, book, hospital, c, out, family, or, when, oklahoma, certified, ohio, number, training, for, like, a, than, be, nursing, ], -, can, writing, patients, no, orthopaedic, attorney, over, ny, mr, “, |

Table 4: Top 100 words influenced by INLP projection (BOW representation, biographies dataset).

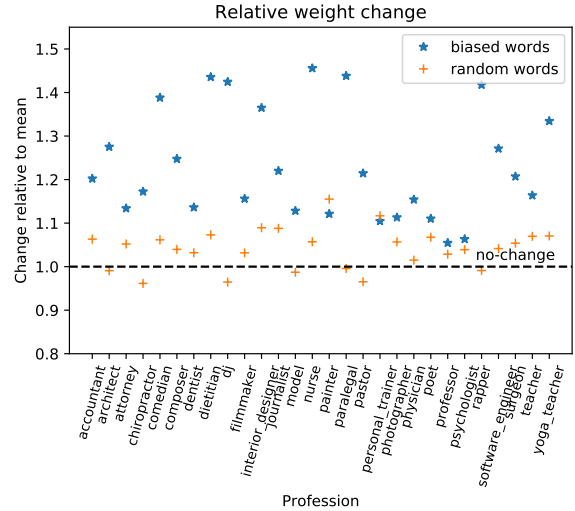


Figure 2: The relative change of biased vs. random words, per profession.

### A.6.2 Bag-of-Word-Vectors Model

In this section, we present an analysis for the influence of INLP projection on the FastText representation of individual words, under the bag-of-word-vectors model, for the biographies experiments (Section 6.3.1). We begin by ordering the vocabulary items by their cosine similarity to each of the top 15 gender directions found in INLP (i.e., their similarity to the weight vector of each classifier). For each gender direction  $w_i$ , we focus on the